Kshitiz Tyagi

Fourth Year Undergraduate
Department of Biological Sciences and Bioengineering

+91-8077616048 | ktyagi23@iitk.ac.in
kshitiz-tyagi | Kshitiztyagi23 | Portfolio

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - 2027	B.Tech	Indian Institute of Technology, Kanpur	8.8/10.0
2022	CBSE(XII)	Dewan Public School, Hapur	95.4%
2020	ICSE(X)	S.D.A Senior Secondary School, Hapur	89.0%

Scholastic Achievements

- Secured the **Academic Excellence Award (2024–2025)** for outstanding academic performance, placing in the top 10 % of the batch.
- Presented a peer-reviewed poster at **CogSci 2026** proposing an Efficiency Hierarchy Model for motion dominance and reaction-time.
- Co-authored a paper identifying developmental trajectories and metacognition in computational thinking across cross-age cohorts.
- Selected as a **Reliance Foundation Scholar** from a pool of 58,000 nationwide applicants based on academic merit and performance.

Work Experience

Data Science Intern | Battery Smart *(May'26 - Jul'26)*

- Engineered an automated feedback pipeline connecting a **PostgreSQL**-backed AI assistant to **Slack** and the **GitHub REST API**, deploying expert-approved knowledge corrections to knowledge base in **3 to 5 seconds**. Eliminated multi-pod race conditions across autoscaled replicas using **atomic database claims**, ensuring zero duplicate commits.
- Architected a continuous, **6-stage data pipeline** to ingest meeting transcripts, preserving spoken institutional knowledge by converting conversations into structured, searchable facts for the AI. Utilized a headless LLM (**Kiro CLI**) to process a **1,613-conversation backlog** at **$0 inference cost**, building a secure AI filter (**MiniLM**) to automatically redact sensitive personal data while operating efficiently within a strict **1GiB memory budget**.
- Authored and SQL-verified over **6,200 lines** of Databricks documentation covering **230+ tables**, successfully eradicating AI hallucinations across 7 business domains. Standardized enterprise diagnostics by developing a 12-page **Root-Cause Analysis (RCA) playbook** equipped with tested SQL patterns for complex metric investigations.
- Resolved training data bias in a **YOLO face-verification model** that incorrectly blocked bearded drivers. Fine-tuned the classifier on CVAT-corrected data and deployed to a live pilot of $\sim$**95,000 images**, eliminating **508 false blocks**, catching **3.3x more genuine image defects**, and boosting model confidence from **0.84 to 0.96**.
- Evaluated a programmatic hybrid retrieval system combining **BM25, MiniLM, and Reciprocal Rank Fusion** to optimize AI routing. Generated a **1,248-node** knowledge graph using **Graphify** to map cross-domain relationships and identify documentation gaps.
- **Autonomous Self-Improving RAG Pipeline:** Engineered an event-driven continuous learning loop via an **MCP (Model Context Protocol)** architecture connecting **PostgreSQL** to Slack and GitHub REST API; auto-committed human-in-the-loop expert corrections into the core retrieval knowledge base in **3–5 seconds** with atomic SQL guarantees preventing multi-pod race conditions.
- **Conversational Knowledge Extraction & Temporal Knowledge Graph:** Architected a 6-stage NLP pipeline extracting structured **Subject-Predicate-Object temporal triplets** from raw meeting transcripts using a headless LLM (**Kiro CLI**); processed a **1,613-conversation backlog** at **$0 inference cost** with local regex PII de-identification and a **MiniLM semantic sensitivity filter** (≥ 0.6 cosine threshold) under a **1GiB** memory budget
- **Hybrid Semantic Retrieval & Topological Graph Routing:** Developed a hybrid retrieval router fusing **Dense Semantic Embeddings (MiniLM)**, **Sparse BM25**, and keyword filters using **Reciprocal Rank Fusion (RRF)** constructed a **1,248-node** / **92-community knowledge graph** via **Graphify (Leiden clustering)** to map cross-domain table dependencies and eliminate context routing blind spots[cite: 3, 5].
- **Domain Adaptation & Hallucination Eradication:** Authored and verified $\sim$**6,200 lines** of structured enterprise schema documentation across **230+ Databricks tables** and 7 business verticals; established a 12-page **RCA prompt diagnostic playbook** with production-tested SQL patterns, completely eliminating LLM query hallucinations.
- **Computer Vision Model Bias Remediation & Fine-Tuning:** Diagnosed severe representation bias in a production **YOLO face-verification classifier** misidentifying facial hair as occlusions; relabeled datasets in **CVAT**, fine-tuned YOLO across 50 epochs, and evaluated over $\sim$**95,000 production images**—eliminating **508 false rejections**, improving genuine defect detection by **3.3x**, and boosting mean confidence from **0.84 to 0.96**

Objective	• Shipped production AI infrastructure for Battery Smart's **Databricks**-scale analytics assistant: a self-correcting feedback loop, a meeting-knowledge extraction pipeline, and a bias fix for a **computer vision** model in verification flow
Approach	• Built a self-correcting feedback loop on top of the AI assistant's **MCP** server: flagged errors get logged to **Postgres**, posted as a Slack review card for approval, then committed to **GitHub** via the REST API and batched into a **PR (Pull Request)**. Prevented replicas from racing on the same correction with one atomic database claim SQL statement. • Meeting transcripts (ingested via API or Slack) go through a **6-stage pipeline**: **PII redaction** first, then a headless **LLM** (**Kiro CLI**) extracts structured knowledge points, stored as vector embeddings within a **1GiB** memory budget. At query time, the assistant matches the user's question against these embeddings to pull in relevant context • Authored ∼6,200 lines of **Databricks** documentation across 230+ tables in 7 business domains, and built a 12-page **RCA** playbook with tested **SQL** patterns; explored a hybrid retrieval system (**BM25**, **MiniLM**, RRF) and a **Graphify** • Diagnosed a training data bias in a **YOLO** face-verification model that repeatedly blocked bearded drivers despite no actual occlusion. Relabeled data in **CVAT** and fine-tuned the model
Results	• Corrections now go from flagged to committed in **3–5 seconds** with zero duplicate commits; the retrained vision model catches **3.3x** more genuine defects with raised confidence scores from **0.84 to 0.96** evaluated upon pilot deployment • The assistant went from no coverage to verified, accurate answers across **7 business domains**, modeled to run at a marginal cost of $0 against a 1,613-conversation backlog • Manual fixes now ship themselves: corrections go live in **3–5 seconds** with zero duplicate commits, and meeting knowledge that used to vanish after each session now gets captured and served back automatically • The retrained vision model passed a **94,924-image** production evaluation before deployment, and the conversational pipeline was demonstrated on real meetings, modeled to run at **$0** marginal cost against a **1,613-conversation** backlog

Machine Learning and AI Systems Engineering Intern. | Skaeo *(May'25 - Jul'25)*

- Integrated a **Traditional Face Liveness Detection System** using **dlib's HOG-based Face Detector** and **68-point Landmark Model** in the **"My RollCall Pro"** app to prevent spoofing. Explored multiple models and implemented an optimized solution using **Blink (EAR)**, **Smile (MAR)**, and **Pupil Movement**, achieving **5 sec Inference Time** post **API Integration**.
- Explored and systematically evaluated multiple **OCR Approaches** across performance benchmarks before selecting and customizing **RapidOCR with ONNX Runtime** for fast and accurate serial number recognition. Developed a complete **Extraction Logic** with **Preprocessing** and **Model Tuning**, achieving **2 sec Inference Time** and high reliability in **Production**.
- Built an **Automated System** using **LangChain** and **Phidata** streamlining **SWOT Analysis, Content Ideation, Generation** and **Post Scheduling**, making marketing campaigns **29.8% More Efficient** and Reducing Manual Work **by 40%**.
- Created a **Telegram Bot** that automatically generates **Contextual Comments** and **Inteligently Reshares Posts** by fetching and analyzing URLs, mimicking **Human-like Ideas**, streamliningtargeted **Content Engagement** and boosting **Interaction**.

Objective	• Engineered ML systems for **real-time security**, **vision**, and **workflow automation** in production environments
Approach	• Devised **Face Liveness Detection** using **dlib** (HOG + 68 landmarks), tracking blink rate, smile, and pupils • Deployed the module into the *My RollCall Pro* app to boost spoof resistance and overall improve ID verification • Integrated **RapidOCR** with **ONNX Runtime** for serial number extraction with preprocessing and handling
Results	• Achieved real-time **5s inference** for face liveness and **2s OCR latency** with high precision in production • Boosted marketing efficiency **by 29.8%**, reduced effort **by 40%**, and ensured **scalability** of deployed systems

RESEARCH PUBLICATIONS & PREPRINTS

When Motion Wins: Cognitive Efficiency in Perceptual Grouping | First Author | Prof. Anveshna Srivastava, IIT Kanpur *(CogSci '26)*

- Designed a psychophysics experiment ($N = 46$, 230 trials) quantifying pairwise cue competition between motion (Common Fate) and 5 classical Gestalt principles.
- Evaluated choice distributions and right-skewed RT latencies using **Exact Binomial Tests**, **Mann-Whitney U**, and **Kruskal-Wallis** non-parametric tests.
- Quantified a graded visual dominance hierarchy for motion cues (**91.3%** vs. Prägnanz, **65.2%** vs. Similarity) and delivered a peer-reviewed **Flash Talk** at **CogSci 2026**.
- Formulated the **Efficiency Hierarchy Model**, conceptualizing perceptual grouping as a cost-sensitive competitive system defined by choice probability and resolution latency.
- Discovered a dissociation between perceptual dominance and efficiency, identifying a 2.7x RT resolution overhead in Good Continuity despite low selection frequency.
- Delivered a peer-reviewed virtual **Flash Talk** and poster at the 48th Annual Meeting of the Cognitive Science Society (**CogSci 2026**, Rio de Janeiro).

Plural Pathways to Computational Thinking | Co-Author | Prof. Anveshna Srivastava, IIT Kanpur *(Ongoing Submission'26)*

- Co-authored an empirical study investigating computational thinking (CT) across **103 participants** (middle school, high school, and educators), pairing a 28-item standardized CT test with hands-on unplugged algorithmic tasks.
- Conducted parametric and non-parametric statistical analyses (**paired t-tests**, **Spearman correlations**, **Cohen's d**); quantified a substantial developmental gap between middle-school students and teachers (**d = 0.99**, $p = 0.009$).
- Demonstrated decoupled correlations ($r = 0.32$–0.39) among test scores, algorithmic output, and metacognitive awareness, providing quantitative evidence that CT operates as a multidimensional construct rather than a single skill.
- Investigated developmental trajectories of algorithmic reasoning across middle-schoolers, high-schoolers, and teachers using standardized assessments alongside physical sorting and insertion paradigms.
- Identified an **expert blind spot** among educators: teachers generated higher-efficiency algorithms but exhibited high variance and low metacognitive articulation of their own underlying logic.

- Formulated empirical recommendations for integrating **metacognitive scaffolding** and unplugged learning frameworks to bridge intuitive physical reasoning into formal computational logic.

KEY PROJECTS

Adaptive Spatial Learning Platform | Prof. Anveshna Srivastava, IIT Kanpur *(Ongoing)*

Objective	• To Design an **adaptive learning platform** for spatial reasoning with interactive 3D tasks & real-time guidance.
Approach	• Building a full-stack platform using **React, FastAPI & PostgreSQL** for sessions, tasks, and data management. • Designing a **real-time feedback** engine to evaluate responses & deliver contextual guidance during tasks. • Prototyping an interactive **3D LEGO-based module** with task validation, scoring, and adaptive prompting. • Plan to **deploy the platform on AWS** using API-driven services, event logging, and normalized data models.
Results	• Targeting a unified learning platform supporting structured sessions, interactive tasks, and real-time feedback. • Designing a modular architecture intended to accommodate additional learning tasks, analytics, and user traffic.

- Building a full-stack adaptive learning platform using **React, FastAPI, & PostgreSQL** for session & data management.
- Designing a **real-time feedback engine** to evaluate learner responses and provide contextual guidance during tasks.
- Prototyping an **interactive 3D LEGO-based learning module** with task validation, scoring, and adaptive prompting.
- Plan to **deploy the platform on AWS** using API-driven services, event logging, production workflows, & normalized data models.
- Structuring an extensible product architecture supporting additional learning modules, analytics, and interaction data.
- Building a **full-stack** adaptive learning platform using **React, FastAPI, and PostgreSQL** to manage sessions and complex user data.
- Implement a **real-time feedback engine** alongside interactive **3D LEGO modules** with task validation and adaptive prompting.
- Designing a **scalable AWS architecture** supporting new learning modules, event logging pipelines, and structured user analytics..

CineScope | CS 661(Big Data Visual Analytics) Course Project | Prof. Soumya Dutta *(May'25 - July'25)*

- Developed an **interactive dashboard** using **Plotly & Dash** to visually analyze global movie data across multiple dimensions
- Derived key patterns by preprocessing **textual and categorical data** for sentiment, genre, and runtime **trend analysis**
- Created **7+ interactive visualizations** with **real-time filters** to uncover insights on budget–revenue trends, genre popularity, company performance, and regional dominance across **800K+ movies**, helps users identify patterns and conclusions
- Built an **interactive Dash & Plotly dynamic visual analytics dashboard** with real-time filters, streamgraphs, and choropleth maps.
- Preprocessed and clustered complex metrics across **900K+ movies** using **PCA and t-SNE** for multidimensional pattern analysis.
- Engineered **7+ interactive visualizations** to analyze box-office revenue trends, genre popularity, and company performance.

Spikeverse | Project Mentor | Brain & Cognitive Science Club, IIT Kanpur *(May'25 - July'25)*

- Efficiently engineered a **Double DQN**-based Atari Breakout agent and now spearheading its conversion into Spiking Network
- Executing structured training pipeline using spike-time encoding, **LIF models**, surrogate gradients, neuromorphic methods
- Projected to cut inference energy by over **40%+** while preserving DQN-level performance across **2,000** training **episodes**

Word For Thought | Project Mentor | Brain & Cognitive Science Club, IIT Kanpur *(Dec'24 - Jan'25)*

- Led the end-to-end development of a speech-based emotion recognition system using **spectrograms** and deep learning
- Converted raw **audio** into spectrograms via **FFT**, trained a **CNN**, and directed a **20+** member team through full pipeline
- Attained **97%** classification accuracy by leveraging time-frequency patterns, enabling real-time and robust emotion detection
- Led a **20+ member team** building an **audio speech pipeline**, converting raw audio into **spectrograms** using **FFT signal processing**.
- Trained a **2D CNN** on extracted time-frequency acoustic patterns, achieving **97% accuracy** for robust real-time **emotion recognition**.

The Minds Mirror | Brain & Cognitive Science Club, IIT Kanpur *(May'24 - Jul'24)*

- Constructed an detailed **EEG dataset** with **200** samples using **PsychoPy** paradigms involving controlled **visual stimuli**
- Applied preprocessing and feature extraction techniques like **FFT, STFT, Autocorrelation, and Welch** for analysis
- Attained **90% Accuracy** on numeric/non-numeric classification using **1D CNN** and augmentation **(SMOTE, Bootstrap)**
- Curated a **200-sample EEG dataset** using **PsychoPy** visual stimuli; extracted robust temporal features using **FFT**, **STFT**, and **Welch**.
- Trained a **1D CNN** with **SMOTE** data augmentation, achieving **90% accuracy** on real-time numeric vs. non-numeric **classification**.

Biobytes | Biosoc, IIT Kanpur *(May'24 - Jun'24)*

- Built a robust DNA classifier using k-mer features and models like **Naive Bayes** and **Random Forest**, achieving accuracy
- Handled zero-frequency issues via **Laplace Smoothing** and used **CountVectorizer**; addressed imbalance and dimensionality

Gan Forge | Project Mentor | Association of Computer Activities, IIT Kanpur *(May'25 - Present)*

- Explored **Visual Recognition and Generation** by building systems for Classification, Detection, and Image Translation
- Trained **DCGAN on curated 60k+ images** to generate high-quality anime faces using deep convolutional architectures
- Trained a Fully Connected Neural Network model for multi-class classification of the **Iris dataset**, achieving **96% accuracy**
- Implemented **CycleGAN** for image translation between **800+ Van Gogh paintings** and **Diverse Natural Images**

Sumo Bot | Robothon Runners-up | Robotics Club, IIT Kanpur *(Apr'24)*

Built and successfully competed with a bot specifically designed for sumo-style robot combat in high-stakes competitive settings

- Led a team of 5 to Design, Prototype, and Test a Durable, **Smartphone-Controlled Bot** with enhanced Maneuverability
- Achieved **2nd place** by optimizing combat strategy, traction control, and precise collision handling in dynamic conditions

Mood Monitor | Brain & Cognitive Science Club, IIT Kanpur *(Dec'23 - Jan'24)*

- Developed a Python-based Stress Detection Model using **NLP and logistic regression** to identify stress indicators in text data
- Performed smart preprocessing and sentiment analysis to significantly boost accuracy, validated using confusion matrix metrics

TECHNICAL SKILLS

- **Languages:** C, C++, Java, Python, SQL **Frameworks:** TensorFlow, PyTorch, FastAPI, React, LangChain
- **ML/CV:** Scikit-learn, YOLO (Ultralytics), dlib, OpenCV, Hugging , RAG, semantics,vector DBs **Data & Libraries:** NumPy, pandas, matplotlib, seaborn, Librosa
- **Platforms/Tools:** AWS, Docker, Databricks, PostgreSQL, Git, Jupyter, PsychoPy, LaTeX, Grafana, Kubernetes

POSITION OF RESPONSIBILITY

Secretary | Brain & Cognitive Science Club, IIT Kanpur *(Jun'23 - Mar'24)*

Induction	• Selected from 300+ applicants for a 2-tier STEM club, contributing to research projects and hackathons
Initiatives	• Conducted interactive sessions on Game Theory and ML for 100+ IITK students using live coding and visuals • Mentored 40+ students across guided projects under the club's research initiative, promoting core learning
Management	• Organized AI/ML lectures, workshops, and guest talks for 200+ students, boosting engagement and learning • Collaborated with Spheron & Innopelux (NVIDIA Inception) to conduct GPU sessions for the community

Secretary | Research Wing, AnC, IIT Kanpur *(Jun'24 - Mar'25)*

Initiatives	• Led sessions on IRS, NSRC, and international internships, including lab tours to C3i Hub and premier facilities • Edited blogs and coordinated senior-led sessions to guide juniors on research journeys
Operations	• Maintained the Research Wing's online and compiled the annual *Gazette* highlighting student / faculty research • Supported event logistics and collaborated with alumni & departments to boost research outreach

Coordinator | BioSoc, IIT Kanpur *(Jun'25 - Mar'26)*

Initiatives	• Spearheaded departmental Research Mentorship Conclaves, orchestrating faculty–student interaction drives that directly facilitated 85+ undergraduate lab attachments across cutting-edge bioengineering domains. • Organized hands-on computational biology workshops, technical lecture series with eminent scientists, and career panels spanning academia, biotechnology industry, and higher studies.
Operations	• Directed a 35+ member multidisciplinary team across technical, events, and marketing verticals, overseeing annual budget allocations, event logistics, and institutional governance. • Boosted departmental student engagement by 60 %, instituted structured peer-mentorship networks, and managed large-scale milestone conferences and community galas.

Research and Projects Secretary| **Biosoc**, IIT Kanpur *(Jun'24–Mar'24)*

Management	• Organized **Prof. Hunt** and engaged **50+ students** through structured **mentorship drives**, fostering connections with **faculty-led research projects** and promoting a culture of academic inquiry and collaboration • Led **Academic events** to promote and foster interest in interdisciplinary research and biotech awareness
Impact	• Fostered student–faculty collaboration and promoted research in biotech which strengthened dept. ecosystem

RELEVANT COURSES

(@ Excellent Grade)(* Online)

Big Data Visual Analytics	Introduction to Machine Learning	Data Structure and Algorithm
Applied Probability and Statistics	Fundamentals of Computing - I	Fundamentals of Computing - II @
Neural Network and DeepLearning*	Convolutional Neural Network*	Supervised Machine Learning*
Principles of Biotechnology	Bioinformatic and Computational Biology @	Introduction to Electronics
Biochemistry @	Organ System Physiology and Anatomy @	Biochemical Engineering @

EXTRA-CURRICULAR ACTIVITIES

- **Secretary, BCS Club:** Led ML sessions for 100+ students and mentored 40+ members across research initiatives *(2023–24)*.
- **Secretary, Research Wing (AnC):** Coordinated lab tours, internship sessions, and compiled the annual research *Gazette (2024–25)*.
- **Coordinator, BioSoc:** Directed *Prof. Hunt* mentorship drives connecting 50+ students to faculty-led research projects *(2024–25)*.
- Won the **Gold Medal** in group dance during Galaxy '25 (Inter-Hall Cultural Festival) at IIT Kanpur.
- Actively volunteered in the annual Blood Donation Camps organized by Raktarpan, IIT Kanpur (2024, 2025).
- Competed in Takneek '24, the Intra-IITK Science and Technology Championship.